

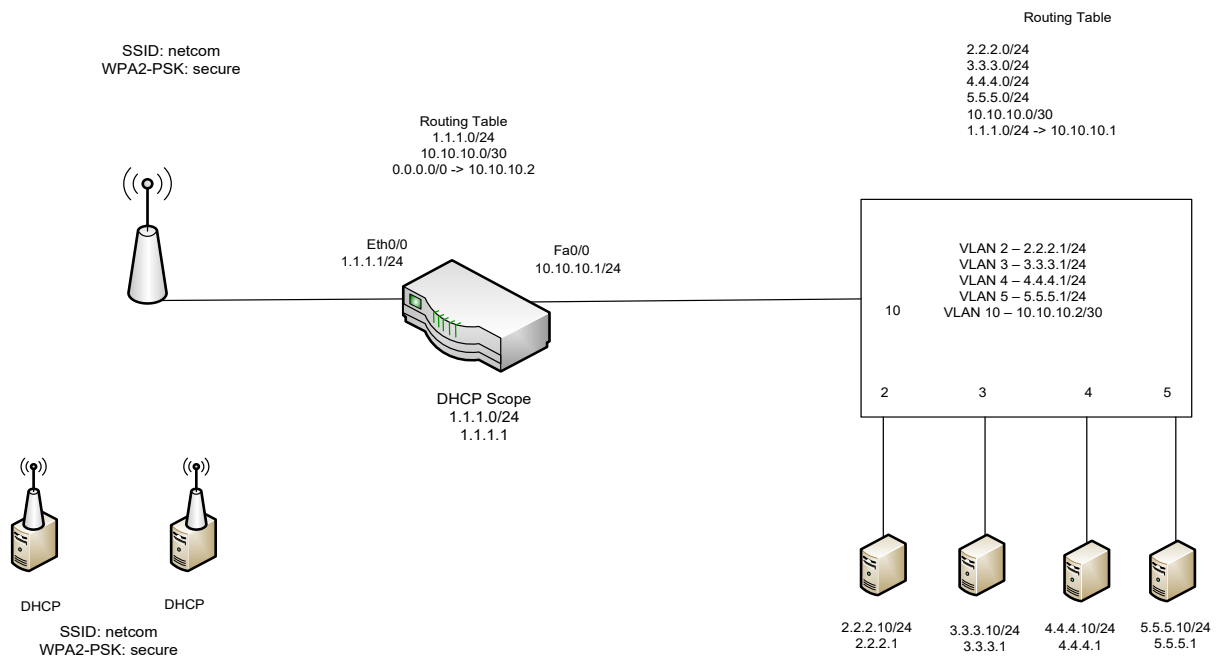
Lab 13

Wireless Networks

Objective:

The purpose of this lab is for the student to understand, create and implement a wireless network. Students will also utilize IOS commands to explore and configure a wireless network using an AP and implementing encryption.

Diagram:



Components:

6 – PC's, 4 with an Ethernet NIC, 2 with wireless NIC's

1 – Layer 3 Switch

1 – Wireless Access Point

1 – 2 Port Router

Procedure:

1. Connect all PC's together on the layer 3 switch and put their static IP addresses, subnet masks and gateways on them. Test connectivity by pinging each machine within the network.
2. Label the Layer 3 switches VLAN IP addresses and ports below

Layer 3 Switch

VLAN	IP	Switch Port
2	2.2.2.1/24	2
3	3.3.3.1/24	3
4	4.4.4.1/24	4
5	5.5.5.1/24	5
10	10.10.10.2/24	10

3. Label the routing tables for the Layer 3 switch and router below.

Router	Layer 3 Switch
C-1.1.1.0/24	C-2.2.2.0/24
C-10.10.10.0/24	C-3.3.3.0/24
S-2.2.2.0/24 -> 10.10.10.2	C-4.4.4.0/24
S-3.3.3.0/24 -> 10.10.10.2	C-5.5.5.0/24
S-4.4.4.0/24 -> 10.10.10.2	C-10.10.10.0/24
S-5.5.5.0/24 -> 10.10.10.2	S-1.1.1.0/24 -> 10.10.10.1

4. Place the switch ports into the proper vlans. Configure all of the switches vlan IP addresses.
5. Enable routing on the Layer 3 switch
6. View the routing table on the Layer 3 switch

7. Verify that the PC's on the switch can ping each other
8. Configure the IP addresses on the router interfaces and turn the interfaces on
9. Place the proper static routes on the Layer 3 switch and the router.
10. On the router create a DHCP pool called net1. Use the 1.1.1.0/24 network and a gateway of 1.1.1.1
 - a. What commands did you enter?
Ip dhcp excluded-address 1.1.1.1 1.1.1.9
Ip dhcp pool net1
Network 1.1.1.0 255.255.255.0
Default-router 1.1.1.1
11. Connect to the access point via a console cable
12. Place an IP address of 1.1.1.2/24 on BVI 1
 - a. What command did you enter?
Interface bvi 1
Ip address 1.1.1.2 255.255.255.0
No shutdown
13. Connect a PC directly to the AP's FastEthernet port
 - a. What kind of cable did you use?
Crossover cable
14. Open a web browser and connect to <http://1.1.1.2>
15. Take a minute to look through the tabs on the left and the options available

- a. Are they the same as the CLI? Yes, though laid out differently.
 - b. Which is easier for you to configure? The GUI
- 16. Under the express settings create a SSID of **open** Leave broadcasting turned on and do not set any type of authentication
- 17. View the wireless networks on one of the wireless PC's?
 - a. Is the open network visible? Yes
 - b. Why? The default setting is broadcasting
- 18. Connect the 2 wireless PC's to the open network.
- 19. Verify the wireless PC's can ping everyone
- 20. Disconnect the 2 wireless PC's from the open network.
- 21. Remove the open network from the AP
- 22. Create a new SSID of **netcom** Turn off broadcasting and enable WPA2 security with AES encryption and a password of **secure**
- 23. View the wireless networks on one of the wireless PC's?
 - a. Is the net network visible? In theory, no, but Packet Tracer did not have that option, so it IS visible...
 - b. Why? The SSID **should** be hidden, but it's not
- 24. Create a manual connect to the netcom network on the 2 wireless PC's.
 - a. What information was needed to setup the network?
The network SSID, the Passkey, and the type of security (WPA2)
- 25. Verify the wireless PC's can ping everyone

- a. Can the non wireless PCs tell the difference between the secure and non secure wireless clients?

They can't tell the difference

- b. What type of wireless network was setup in this lab?

This was an example of 802.11n wireless network – infrastructure.

26. Create the lab diagram below (PCs, router and switch connections) in Packet Tracer or Visio. **Label the IP addresses, subnet masks, and gateways** used by each device. **Label the routing tables next to the switch and router** and the **switches vlan IP addresses and port vlan information**. Label the wireless SSID and keys next to the AP and wireless clients. Label the router DHCP scope.

